**Solutions Tutorial 1**

1. Consider functions $g:\mathbb{R}^2\to\mathbb{R}$, $f:\mathbb{R}\to\mathbb{R}$ which satisfy the relationship

$$g(x,y)=x+y+f(x-y).$$

Given that $g(x,0)=x^2$. Determine the explicit forms of the functions $f$ and $g$.

Solution. We substitute $g(x,0)=x^2$ to obtain $x^2=z=x+0+f(x-0)$. Therefore, $f(x)=x^2-x$ and thus $f(x-y)=(x-y)^2-(x-y)$. Using this we get

$$g(x,y)=x+y+(x-y)^2-(x-y)=2y+(x-y)^2.$$

2. Determine $g(x)$ for $x>0$ if $g\left(\frac{y}{x}\right)=\frac{\sqrt{x^2+y^2}}{x}, x>0$.

Solution. Let $u=\frac{y}{x}$ and write $g(u)=g\left(\frac{y}{x}\right)=\frac{\sqrt{x^2+y^2}}{x}=\sqrt{1+\left(\frac{y}{x}\right)^2}=\sqrt{1+u^2}$. Thus, in term of $x$: $g(x)=\sqrt{1+x^2}$.

3. Find the domains for the following functions. Sketch them as regions on the plane.

(a) $f(x,y)=\frac{1}{x}\sqrt{1-y}$

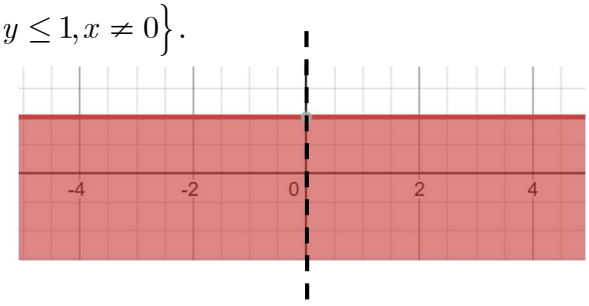
(b) $g(x,y)=\ln(5-x^2-y^2)$

(c) $h(x,y)=\frac{\ln(1-x^2)}{xy}$

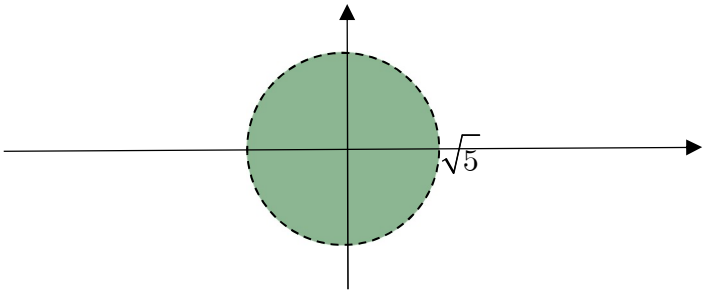
(d) $q(x,y)=\sqrt{(1-x^2)(1-y^2)}$

Solution.

(a) $D=\left\{(x,y)\in\mathbb{R}^2 : y\le 1, x\neq 0\right\}$.



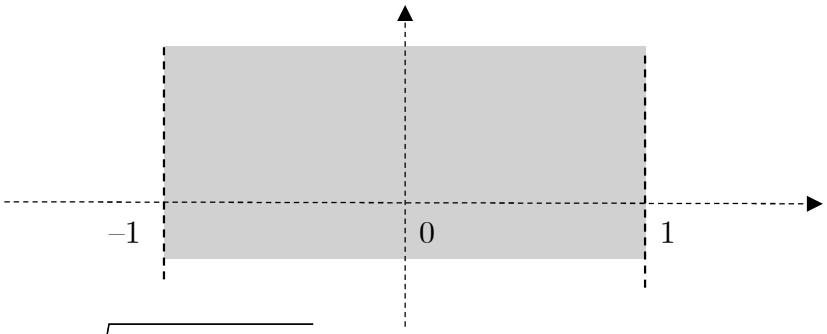

(b) Given $g(x,y)=\ln(5-x^2-y^2)$. Then $D=\{(x,y)\in\mathbb{R}^2 : x^2+y^2<5\}$. Points inside a circle of radius $\sqrt{5}$ centered at the origin.



(c) Given $h(x,y)=\frac{\ln(1-x^2)}{xy}$. The divisions are defined when $x\neq 0$ and $y\neq 0$. The logarithm is defined when $1-x^2>0$ which means $x^2<1$ or equivalently $|x|<1$. So,

$$D=\{(x,y)\in\mathbb{R}^2:|x|<1, x,y\neq 0\}.$$

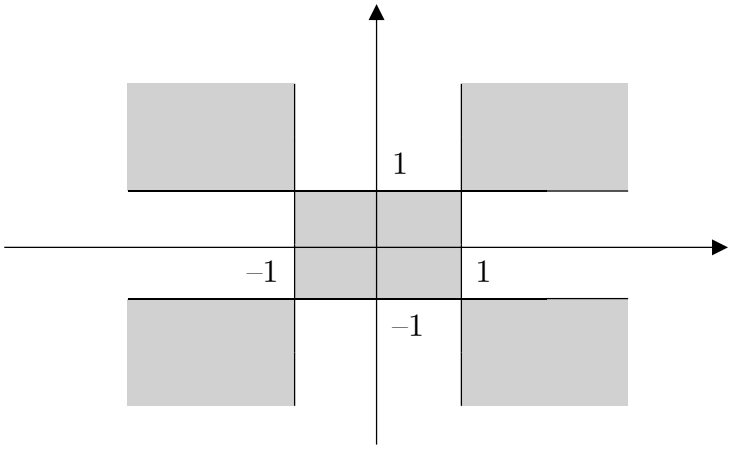
This is a strip of width 2 stretched along the $y$-axis with the points on the lines $x=0$ and $y=0$ removed.




(d) Given $q(x,y)=\sqrt{(1-x^2)(1-y^2)}$. For the square root to be defined we need $(1-x^2)(1-y^2)\geq 0$. This means that the expressions $(1-x^2)$, $(1-y^2)$ are either both non-negative or both non-positive. In the first case we have the region $x^2\leq 1$, $y^2\leq 1$ or equivalently $|x|\leq 1$, $|y|\leq 1$. This region looks like a square around the origin. In the second case we have $x^2\geq 1$, $y^2\geq 1$ or equivalently $|x|\geq 1$, $|y|\geq 1$. This region looks like four infinite quarter-planes. The whole domain can thus be written as a union of these two regions:

$$D=\{(x,y)\in\mathbb{R}^2:|x|\leq 1,|y|\leq 1\}\cup\{(x,y)\in\mathbb{R}^2:|x|\geq 1,|y|\geq 1\}.$$

It contains a square around the origin with a side of length 2 and four infinite quarter-planes.



4. Consider the function $F:\mathbb{R}^2\to\mathbb{R}$ defined by

$$F(x,y)=\begin{cases}\frac{x^2}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}.$$

Find $\lim\limits_{(x_n,y_n)\to(0,0)}F(x,y)$ for (a) $(x_n,y_n)=(0,\frac{1}{n})$; (b) $(x_n,y_n)=(\frac{1}{n},0)$; (c) $(x_n,y_n)=(\frac{1}{n},\frac{2}{n})$ .

Is $F(x,y)$ continuous at $(x,y)=(0,0)$?

Solution. Note that $(0,\frac{1}{n})\to(0,0)$, $(\frac{1}{n},0)\to(0,0)$, $(\frac{1}{n},\frac{2}{n})\to(0,0)$ as $n\to\infty$.

(a) $\lim\limits_{(x,y)\to(0,0)}F(x,y)=\lim\limits_{n\to\infty}F(0,\frac{1}{n})=\lim\limits_{n\to\infty}\frac{0}{0+1/n^2}=\lim\limits_{n\to\infty}0=0$.

(b) $\lim\limits_{(x,y)\to(0,0)}F(x,y)=\lim\limits_{n\to\infty}F(\frac{1}{n},0)=\lim\limits_{n\to\infty}\frac{1/n^2}{1/n^2+0}=\lim\limits_{n\to\infty}1=1$.

(c) $\lim\limits_{(x,y)\to(0,0)}F(x,y)=\lim\limits_{n\to\infty}F(\frac{1}{n},\frac{2}{n})=\lim\limits_{n\to\infty}\frac{1/n^2}{1/n^2+4/n^2}=\lim\limits_{n\to\infty}\frac{1}{5}=\frac{1}{5}$.

Limits in (a), (b) and (c) are all different. Thus, $\lim\limits_{(x,y)\to(0,0)}F(x,y)$ does not exist and so $F$ is not continuous at $(x,y)=(0,0)$.

Alternative. Consider pathways $y=x$ and $y=2x$, then

$$\lim_{(x,y)\to(0,0)}F(x,y)=\lim_{x\to0}F(x,x)=\lim_{x\to0}\frac{1}{2}=\frac{1}{2}\neq\frac{1}{5}=\lim_{x\to0}\frac{1}{5}=\lim_{x\to0}F(x,2x)=\lim_{(x,y)\to(0,0)}F(x,y).$$

5. Consider a function $G(x,y)=\frac{x-y}{x+y}$ defined for $x\neq-y$. Find $\lim\limits_{(x_n,y_n)\to(0,0)}G(x,y)$ for

(a) $(x_n,y_n)=(0,\frac{1}{n})$ (b) $(x_n,y_n)=(\frac{1}{n},0)$ (c) $(x_n,y_n)=(\frac{1}{n},\frac{1}{n})$

Can $G$ be defined at (0,0) so that it is continuous at that point?

Solution. Note that $(0,\frac{1}{n})\to(0,0)$, $(\frac{1}{n},0)\to(0,0)$, $(\frac{1}{n},\frac{1}{n})\to(0,0)$ as $n\to\infty$.

(a) $\lim\limits_{(x,y)\to(0,0)}G(x,y)=\lim\limits_{n\to\infty}G(0,\frac{1}{n})=\lim\limits_{n\to\infty}\frac{0-1/n}{0+1/n}=\lim\limits_{n\to\infty}(-1)=-1$.

(b) $\lim\limits_{(x,y)\to(0,0)}G(x,y)=\lim\limits_{n\to\infty}G(\frac{1}{n},0)=\lim\limits_{n\to\infty}\frac{1/n-0}{1/n+0}=\lim\limits_{n\to\infty}1=1$.

(c) $\lim\limits_{(x,y)\to(0,0)}G(x,y)=\lim\limits_{n\to\infty}G(\frac{1}{n},\frac{1}{n})=\lim\limits_{n\to\infty}\frac{1/n-1/n}{1/n+1/n}=\lim\limits_{n\to\infty}0=0$.

The limit obtained in (a), (b) and (c) are all different. Thus, $\lim\limits_{(x,y)\to(0,0)}G(x,y)$ does not exist and so $G$ is not continuous and cannot be made continuous at (0,0).

Alternative. Consider pathways $y=x$ and $y=2x$, then

$$\lim_{(x,y)\to(0,0)}G(x,y)=\lim_{x\to0}G(x,x)=0\neq-\frac{1}{3}=\lim_{x\to0}G(x,2x)=\lim_{(x,y)\to(0,0)}G(x,y).$$

6. Consider a function $H(x,y)=\frac{x^2+2y}{x-y}$ defined for $x\neq y$. Can the value of the function $H$ be defined at $(x,y)=(0,0)$ so that it is continuous at that point?

Solution. Consider $(0,\frac{1}{n}),(\frac{1}{n},0)\to(0,0)$ as $n\to\infty$.

$\lim\limits_{(x,y)\to(0,0)} H(x,y)=\lim\limits_{n\to\infty} H(0,\frac{1}{n})=\lim\limits_{n\to\infty}\frac{0+2/n}{0-1/n}=\lim\limits_{n\to\infty}(-2)=-2$; but not equal to

$\lim\limits_{(x,y)\to(0,0)} H(x,y)=\lim\limits_{n\to\infty} H(\frac{1}{n},0)=\lim\limits_{n\to\infty}\frac{1/n^2}{1/n}=\lim\limits_{n\to\infty}\frac{1}{n}=0$.

Thus, $\lim\limits_{(x,y)\to(0,0)} H(x,y)$ does not exist and so $H$ is not continuous and cannot be made continuous at (0,0).

Alternative. Consider pathways $y=-\frac{1}{2}x^2$ and $y=-x$, then

$$\lim_{x\to0} H(x,-\tfrac{1}{2}x^2)=0\neq-1=\lim_{x\to0} H(x,-x).$$

7. Find the following limits if they exist; or justify why their limits do not exist.

(a) $\lim\limits_{(x,y)\to(2,-1)}\frac{2x+3y}{4x-3y}$ (b) $\lim\limits_{(x,y)\to(5,-2)}\sqrt[3]{\frac{x^2-y}{y^2+x-1}}$ (c) $\lim\limits_{(x,y)\to(1,0)}\frac{4xy^2}{x^2+3y^4}$ (d) $\lim\limits_{(x,y)\to(0,0)}\frac{4xy^2}{x^2+3y^4}$

Solution. (a) $\lim\limits_{(x,y)\to(2,-1)}\frac{2x+3y}{4x-3y}=\frac{\lim\limits_{(x,y)\to(2,-1)}(2x+3y)}{\lim\limits_{(x,y)\to(2,-1)}(4x-3y)}=\frac{2\times\lim\limits_{x\to2}x+3\times\lim\limits_{y\to-1}y}{2\times\lim\limits_{x\to2}x-3\times\lim\limits_{y\to-1}y}=\frac{4-3}{8+3}=\frac{1}{11}$

(b) $\lim\limits_{(x,y)\to(5,-2)}\sqrt[3]{\frac{x^2-y}{y^2+x-1}}=\sqrt[3]{\frac{25+2}{4+5-1}}=\frac{3}{2}$ (c) $\lim\limits_{(x,y)\to(1,0)}\frac{4xy^2}{x^2+3y^4}=0$

(d) Let $f(x,y)=\frac{4xy^2}{x^2+3y^4}$. Note that $\left(\frac{1}{n^2},\frac{1}{n}\right)\to\left(0,0\right)$ and $\left(\frac{1}{n},0\right)\to\left(0,0\right)$ as $n\to\infty$. However, $\lim\limits_{n\to\infty} f\left(\frac{1}{n^2},\frac{1}{n}\right)=\lim\limits_{n\to\infty}\frac{4/n^4}{1/n^4+3/n^4}=1\neq0=\lim\limits_{n\to\infty} f\left(\frac{1}{n},0\right)$. Therefore, the limit does not exist at (0, 0).

Alternative. Consider $x=y^2$ and $y=x$, then $\lim\limits_{y\to0} f(y^2,y)=1\neq0=\lim\limits_{y\to0} f(y,y)$.

8. Define $G(x,y)=\begin{cases}\frac{(x-1)^2}{(x-1)^2+y^2} & (x,y)\neq(1,0)\\ 0 & (x,y)=(1,0)\end{cases}$. Evaluate $\lim\limits_{(x,y)\to(1,0)} G(x,y)$ along the line

(i) $y=0$ (ii) $x=1$ (iii) $x-1=y$.

Hence, determine whether $G$ is continuous at (1, 0).

Solution. Note that (i) $\lim\limits_{x\to1} G(x,0)=\lim\limits_{x\to1}\frac{(x-1)^2}{(x-1)^2}=1$; (ii) $\lim\limits_{y\to0} G(1,y)=\lim\limits_{y\to0}\frac{0}{0+y^2}=0$; and (iii) $\lim\limits_{y\to0} G(y+1,y)=\lim\limits_{y\to0}\frac{y^2}{2y^2}=\frac{1}{2}$. All of the above answers are different, which means $\lim\limits_{(x,y)\to(1,0)} G(x,y)$ does not exist at (1, 0). Hence, $G$ is not continuous at (1, 0).

9. Let $f(x,y) = x^2 - y^2$ defined on the whole of $\mathbb{R}^2$. Express this function in terms of the polar coordinates $(r,\theta)$. If the distance of $(x,\ y)$ from the origin is equal to 1, what are the largest and smallest values of $f$?

Solution. Let $x = r\cos\theta$, $y = r\sin\theta$. Then $f(r,\theta) = r^2\left(\cos^2\theta - \sin^2\theta\right) = r^2\cos 2\theta$. Given $r = 1$ (distance from the origin), so $f(1,\theta) = f(\theta) = \cos 2\theta$. Hence, $f_{\min} = -1$ and $f_{\max} = 1$.

10. Evaluate $\lim\limits_{(x,y)\to(0,0)} \frac{x^4+y^4}{x^2+y^2}$.

Solution. Let $x = r\cos\theta, y = r\sin\theta$. Then $\frac{x^4+y^4}{x^2+y^2} = \frac{r^4(\cos^4\theta+\sin^4\theta)}{r^2(\cos^2\theta+\sin^2\theta)} = r^2(\cos^4\theta + \sin^4\theta)$. So, $\lim\limits_{(x,y)\to(0,0)} \frac{x^4+y^4}{x^2+y^2} = \lim\limits_{r\to 0^+} r^2(\cos^4\theta + \sin^4\theta) = 0$ as we apply the Squeeze theorem, i.e., $0 < r^2\left(\cos^4\theta + \sin^4\theta\right) < 2r^2$ for all $\theta$ and $\lim\limits_{r\to 0^+} 0 = 0 = \lim\limits_{r\to 0^+}(2r^2)$.

Remark: To be exact, $\frac{1}{2} \le \cos^4\theta + \sin^4\theta \le 1$ as explained below.

$$\begin{aligned}\sin^4\theta + \cos^4\theta &= \sin^4\theta + 2\sin^2\theta\cos^2\theta + \cos^4\theta - 2\sin^2\theta\cos^2\theta \\ &= (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta = 1 - \tfrac{1}{2}\sin^2 2\theta\end{aligned}$$

The result follows since $0 \le \sin^2 2\theta \le 1$.

Alternative. We first note that the Squeeze theorem for two-variable functions work just like the single variable version, i.e., let $\delta > 0$ and suppose $g(x,y) \le f(x,y) \le h(x,y)$ such that $\lim\limits_{(x,y)\to(a,b)} g(x,y) = \lim\limits_{(x,y)\to(a,b)} h(x,y) = L$ for all $(x,y)$ with $0 < \|(x,y)-(a,b)\| < \delta$. Then $\lim\limits_{(x,y)\to(a,b)} f(x,y) = L$.

Now since $(x^2+y^2)^2 = x^4 + 2x^2y^2 + y^4 \ge x^4 + y^4 \Rightarrow 0 < \frac{x^4+y^4}{x^2+y^2} \le x^2 + y^2$, it follows that $\lim\limits_{(x,y)\to(0,0)} \frac{x^4+y^4}{x^2+y^2} = 0$ because $\lim\limits_{(x,y)\to(0,0)} 0 = 0 = \lim\limits_{(x,y)\to(0,0)} (x^2+y^2)$.

11. Prove that $\lim\limits_{(x,y)\to(0,0)} \left(\frac{x^2-y^2}{x^2+y^2}\right)$ does not exist.

Proof. $(0,\frac{1}{n}),(\frac{1}{n},0) \to (0,0)$ as $n\to\infty$, $\lim\limits_{(x,y)\to(0,0)} \left(\frac{x^2-y^2}{x^2+y^2}\right) = \lim\limits_{n\to\infty}\left(\frac{0-1/n^2}{0+1/n^2}\right) = -1$, but $\lim\limits_{(x,y)\to(0,0)} \left(\frac{x^2-y^2}{x^2+y^2}\right) = \lim\limits_{n\to\infty}\left(\frac{1/n^2-0}{1/n^2+0}\right) = 1$. So, the limit does not exist.

Alternative. Consider $x = 2y$ and $y = x$, then $\lim\limits_{y\to 0} f(2y,y) = \frac{3}{5} \ne 0 = \lim\limits_{y\to 0} f(y,y)$.

**Extra Exercises**

12. Evaluate, if possible. Otherwise, state the reason why the limit does not exist.

(a) $\lim\limits_{(x,y)\to(2,-1)}\left(xy+x^2\right)$ (b) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x}{x^2+y^2}\right)$ (c) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^2}{x^2+y^2}\right)$

(d) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^3}{x^2+y^2}\right)$ (e) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^2y^2}{x^2+y^4}\right)$ (f) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{(e^y-1)\sin x}{xy}\right)$

Solution. (a) $\lim\limits_{(x,y)\to(2,-1)}\left(xy+x^2\right)=2(-1)+4=2$

(b) $(0,\frac{1}{n})\to(0,0)$ as $n\to\infty$, $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x}{x^2+y^2}\right)=\lim\limits_{n\to\infty}\left(\frac{0}{0+1/n^2}\right)=0$ but $(\frac{1}{n},0)\to(0,0)$ as $n\to\infty$, $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x}{x^2+y^2}\right)=\lim\limits_{n\to\infty}\left(\frac{1/n}{1/n^2+0}\right)=\infty$. So, the limit does not exist at (0,0).

(c) $(0,\frac{1}{n})\to(0,0)$ as $n\to\infty$, $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^2}{x^2+y^2}\right)=\lim\limits_{n\to\infty}\left(\frac{0}{0+1/n^2}\right)=0$ but $(\frac{1}{n},0)\to(0,0)$ as $n\to\infty$, $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^2}{x^2+y^2}\right)=\lim\limits_{n\to\infty}\left(\frac{1/n^2}{1/n^2+0}\right)=1$. So, the limit does not exist at (0,0).

(d) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^3}{x^2+y^2}\right)=\lim\limits_{r\to0^+}\left(\frac{(r\cos\theta)^3}{r^2}\right)=0$ since $\left|\cos\theta\right|\le1$ for all $\theta$.

(e) Since $\frac{x^2y^2}{x^2+y^4}=\frac{r^4(\sin\theta\cos\theta)^2}{r^2\cos^2\theta+r^4\sin^4\theta}=\frac{(r\sin\theta\cos\theta)^2}{\cos^2\theta+r^2\sin^4\theta}=r^2\sin^2\theta\frac{\cos^2\theta}{\cos^2\theta+r^2\sin^4\theta}$, it follows that

$$\lim_{(x,y)\to(0,0)}\left(\frac{x^2y^2}{x^2+y^4}\right)=\lim_{r\to0^+}\left(r^2\sin^2\theta\frac{\cos^2\theta}{\cos^2\theta+r^2\sin^4\theta}\right)=\lim_{r\to0^+}\left(r^2\sin^2\theta\right)\times\lim_{r\to0^+}\left(\frac{\cos^2\theta}{\cos^2\theta+r^2\sin^4\theta}\right)=0$$

because $0\le\sin^2\theta\le1$ and $0\le\frac{\cos^2\theta}{\cos^2\theta+r^2\sin^4\theta}\le\frac{\cos^2\theta}{\cos^2\theta}=1$ for all $\theta$ (apply the Squeeze theorem on both limits.)

(f) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{(e^y-1)\sin x}{xy}\right)=\left(\lim\limits_{x\to0}\frac{\sin x}{x}\right)\left(\lim\limits_{y\to0}\frac{e^y-1}{y}\right)=1\times\lim\limits_{y\to0}\frac{e^y}{1}=1$ (applying L' Hôpital's rule).

Remark: Suppose $F(x,y):\mathbb{R}^2\to\mathbb{R}$ is a separable function, i.e., $F(x,y)=f(x)g(y)$. Then $\lim\limits_{(x,y)\to(a,b)}F(x,y)=\lim\limits_{x\to a}f(x)\times\lim\limits_{y\to b}g(y)$ if both $\lim\limits_{x\to a}f(x)$ and $\lim\limits_{y\to b}g(y)$ exist.

13. Evaluate each of the following limits or show that it does not exist.

(a) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^3-y^3}{x^2+y^2}\right)$ (b) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^2-y^4}{x^2+y^4}\right)$

Solution. (a) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{x^3-y^3}{x^2+y^2}\right)=\lim\limits_{r\to0^+}\left(\frac{r^3(\cos^3\theta-\sin^3\theta)}{r^2}\right)=\lim\limits_{r\to0^+}\left(r(\cos^3\theta-\sin^3\theta)\right)=0$ because $\left|\cos^3\theta-\sin^3\theta\right|\le\left|\sin^3\theta\right|+\left|\cos^3\theta\right|\le2$ and applying the Squeeze theorem.

(b) $(0,\frac{1}{n}),(\frac{1}{n},0)\to(0,0)$ as $n\to\infty$,

$$\lim_{(x,y)\to(0,0)}\left(\frac{x^2-y^4}{x^2+y^4}\right)=\lim_{n\to\infty}\left(\frac{0-1/n^4}{0+1/n^4}\right)=-1\ne1=\lim_{n\to\infty}\left(\frac{1/n^2-0}{1/n^2+0}\right)=\lim_{(x,y)\to(0,0)}\left(\frac{x^2-y^4}{x^2+y^4}\right).$$

So, limit does not exist at (0,0).

14. Compute the following limits or explain why they do not exist.

(a) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{xy}{x^2+y^2}\right)$ (b) $\lim\limits_{(x,y)\to(0,0)}\left(\frac{\sin(xy)}{x^2+y^2}\right)$

Solution. (a) Consider $(0,\frac{1}{n}),(\frac{1}{n},\frac{1}{n})\to(0,0)$ as $n\to\infty$, $\lim\limits_{(x,y)\to(0,0)}\left(\frac{xy}{x^2+y^2}\right)=\lim\limits_{n\to\infty}\left(\frac{0}{0+1/n^2}\right)=0$ but $\lim\limits_{(x,y)\to(0,0)}\left(\frac{xy}{x^2+y^2}\right)=\lim\limits_{n\to\infty}\left(\frac{1/n^2}{1/n^2+1/n^2}\right)=\frac{1}{2}$. So, the limit does not exist at (0,0).

(b) Suppose $\lim\limits_{(x,y)\to(0,0)}\left(\frac{\sin(xy)}{x^2+y^2}\right)=\lim\limits_{(x,y)\to(0,0)}\left(\frac{xy}{x^2+y^2}\right)\times\lim\limits_{(x,y)\to(0,0)}\left(\frac{\sin(xy)}{xy}\right)$ exists. Since $\lim\limits_{(x,y)\to(0,0)}\frac{\sin(xy)}{xy}=\lim\limits_{u\to0}\frac{\sin u}{u}=1$, it follows that $\lim\limits_{(x,y)\to(0,0)}\left(\frac{xy}{x^2+y^2}\right)$ exists.

But this is a contradiction from (a), thus $\lim\limits_{(x,y)\to(0,0)}\left(\frac{\sin(xy)}{x^2+y^2}\right)$ does not exist.

15. Evaluate each of the following limits $\lim\limits_{(x,y)\to(0,0)}f(x,y)$ or show that it does not exist.

(a) $f(x,y)=\begin{cases}\frac{x^2}{y-x}, y\neq x\\ 0, \quad y=x\end{cases}$ (b) $f(x,y)=\begin{cases}\frac{y^8}{y-x}, y\neq x\\ 0, \quad y=x\end{cases}$

Solution. (a) Consider the pathways $y=x\pm x^2$, we have

$$\lim_{x\to0}f(x,x+x^2)=1\neq-1=\lim_{x\to0}f(x,x-x^2).$$

Thus, limit does not exist at (0,0).

(b) Consider the pathway $y=x\pm x^8$, then

$$\lim_{x\to0}f(x,x+x^8)=1\neq-1=\lim_{x\to0}f(x,x-x^8).$$

Thus, limit does not exist at (0,0).

**Past years test/final exam questions**

16. Let $G(x,y)=\frac{x+y}{x^2+y}$.

(a) Write the domain of $G(x,y)$. (b) Does $\lim\limits_{(x,y)\to(1,-1)} G(x,y)$ exist?

(c) Determine whether $G(x,y)$ can be made continuous at $(1,-1)$ or not.

Solution. (a) Domain of $G=\left\{(x,y)\in\mathbb{R}^2 : x^2+y\neq 0\right\}$.

(b) Consider pathway $x=-y$, $\lim\limits_{(x,y)\to(1,-1)} \frac{x+y}{x^2+y}=\lim\limits_{y\to -1}\frac{0}{y^2+y}=0$.

Consider pathway $y=x-2$, $\lim\limits_{(x,y)\to(1,-1)} \frac{x+y}{x^2+y}=\lim\limits_{x\to 1}\frac{2x-2}{x^2+x-2}=\lim\limits_{x\to 1}\frac{2}{x+2}=\frac{2}{3}$.

Since $0\neq\frac{2}{3}$, it follows that $\lim\limits_{(x,y)\to(1,-1)} G(x,y)$ does not exist

(c) It is not possible to have $G(x,y)$ continuous at $(1,-1)$.

17. Let $f(x,y)=\begin{cases}\frac{2y^2}{2x-y}, & x\neq y\\ 0, & x=y\end{cases}$. Show that $f(x,y)$ is not continuous at $(0,0)$.

Proof. Use $y=0$ and $x=y^2+\frac{1}{2}y$: $\lim\limits_{x\to 0} f(x,0)=0\neq 1=\lim\limits_{y\to 0}\frac{2y^2}{2y^2}=\lim\limits_{y\to 0} f(y^2+\frac{1}{2}y,y)$.

Hence this limit does not exist at $(0,0)$, and $f(x,y)$ is not continuous at $(0,0)$.

18. Determine whether $G(x,y)=\begin{cases}\frac{\sqrt{2}(x-1)}{x+y-1} & (x,y)\neq(1,0)\\ 0 & (x,y)=(1,0)\end{cases}$ is continuous at $(1,0)$.

Solution. $\left(1,\frac{1}{n}\right),\left(1+\frac{1}{n},0\right)\to(1,0)$ as $n\to\infty$, but $\lim\limits_{n\to\infty} G\left(1,\frac{1}{n}\right)=0$ is not equal to $\lim\limits_{n\to\infty} G\left(1+\frac{1}{n},0\right)=\sqrt{2}\lim\limits_{n\to\infty}\left(\frac{1+\frac{1}{n}-1}{1+\frac{1}{n}+0-1}\right)=\sqrt{2}$. So, $\lim\limits_{(x,y)\to(1,0)} G(x,y)$ does not exist and thus it is not continuous at $(1,0)$.

Alternative. Consider $y=x-1$ and $x=1$. Then $\lim\limits_{x\to 1} G\left(x,x-1\right)=\frac{1}{\sqrt{2}}\neq 0=\lim\limits_{y\to 0} G\left(1,y\right)$.

19. Let $f:\mathbb{R}^2 \to \mathbb{R}$ be a function given by $f(x,y)=\frac{x^2y}{x^4+2y^2}$. Prove that

(a) $f$ cannot be made continuous at $(0,\,0)$ (b) $f$ is differentiable at $\mathbb{R}^2 \setminus \{(0,0)\}$.

Proof. (a) Consider $(\frac{1}{n},0),(\frac{1}{n},\frac{1}{n^2}) \to (0,0)$ as $n \to \infty$. Now

$$\lim_{n\to\infty} f(\tfrac{1}{n},0)=0 \text{ is different from } \lim_{n\to\infty} f(\tfrac{1}{n},\tfrac{1}{n^2})=\frac{\frac{1}{n^4}}{\frac{1}{n^4}+\frac{2}{n^4}}=\frac{1}{3}.$$

So, the $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist and so $f(x,y)=\frac{x^2y}{x^4+2y^2}$ cannot be made continuous at $(0,0)$.

(b) We see that both partial derivatives $f_x=\frac{2xy(x^4+2y^2)-x^2y(4x^3)}{(x^4+2y^2)^2}$ and $f_y=\frac{x^2(x^4+2y^2)-x^2y(4y)}{(x^4+2y^2)^2}$ exist on $\mathbb{R}^2 \setminus \{(0,0)\}$. Since $f_x$ and $f_y$ are continuous for all points on $\mathbb{R}^2 \setminus \{(0,0)\}$ (Theorem 1.9), it follows that, by Theorem 2.18, $f$ is continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$.